

REPORT

Travelling Salesman Problem (TSP)

Aim

To find the shortest possible route that visits every given city exactly once and returns to the starting city while minimizing the total travel distance or cost.

Objectives

- To understand the Travelling Salesman Problem.
- To find an optimal route among multiple cities.
- To minimize the total travelling cost or distance.
- To understand optimization techniques used in AI.

Requirements

- Computer/Laptop
- Programming language such as C, C++, Java, or Python
- Code editor/IDE
- Set of cities and their distance/cost matrix

Theory

The **Travelling Salesman Problem (TSP)** is a well-known optimization problem in which a salesman must visit a set of cities exactly once and return to the starting city. The objective is to find the route with the **minimum total distance or cost**. TSP is a **combinatorial optimization problem** and can be solved using techniques such as brute force, dynamic programming, branch and bound, and heuristic methods.

Algorithm

1. Start with the given set of cities.
2. Select a starting city.
3. Generate possible routes visiting every city exactly once.
4. Calculate the total distance or cost of each route.
5. Compare the total costs of all possible routes.
6. Select the route with the minimum total cost.
7. Return to the starting city.
8. Display the optimal route and its total distance/cost.

PROGRAM

```
from itertools import permutations
```

```
graph = [  
    [0, 10, 15, 20],  
    [10, 0, 35, 25],  
    [15, 35, 0, 30],  
    [20, 25, 30, 0]  
]
```

```
n = len(graph)  
cities = list(range(1, n))
```

```
min_cost = float('inf')
```

```
for perm in permutations(cities):  
    cost = 0  
    current = 0
```

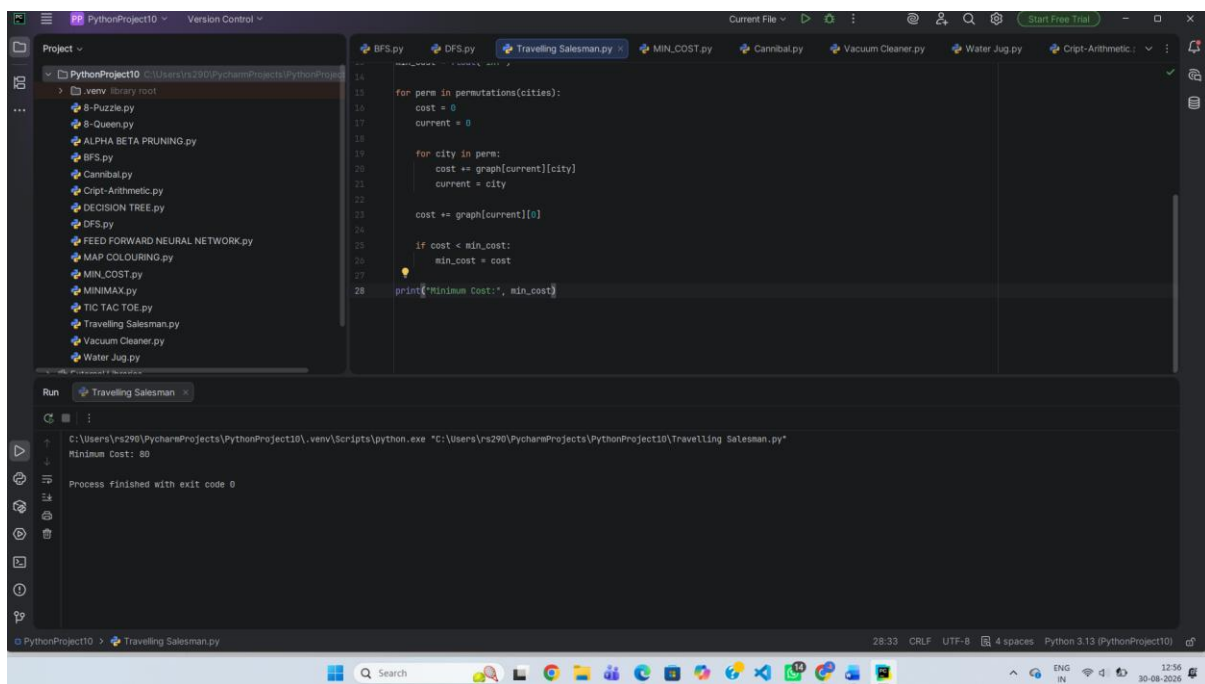
```
for city in perm:  
    cost += graph[current][city]  
    current = city
```

```
cost += graph[current][0]
```

```
if cost < min_cost:  
    min_cost = cost
```

```
print("Minimum Cost:", min_cost)
```

OUTPUT



```
PythonProject10  
├── .venv  
├── 8-Puzzle.py  
├── 8-Queen.py  
├── ALPHA BETA PRUNING.py  
├── BFS.py  
├── Cannibal.py  
├── Crypt-Arithmetic.py  
├── DECISION TREE.py  
├── DFS.py  
├── FEED FORWARD NEURAL NETWORK.py  
├── MAP COLOURING.py  
├── MIN_COST.py  
├── MINIMAX.py  
├── TIC TAC TOE.py  
├── Travelling Salesman.py  
├── Vacuum Cleaner.py  
└── Water Jug.py
```

```
Travelling_Salesman.py  
14  
15 for perm in permutations(cities):  
16     cost = 0  
17     current = 0  
18  
19     for city in perm:  
20         cost += graph[current][city]  
21         current = city  
22  
23     cost += graph[current][0]  
24  
25     if cost < min_cost:  
26         min_cost = cost  
27  
28 print("Minimum Cost:", min_cost)
```

```
Run  
C:\Users\rs290\PycharmProjects\PythonProject10\.venv\Scripts\python.exe "C:\Users\rs290\PycharmProjects\PythonProject10\Travelling_Salesman.py"  
Minimum Cost: 80  
Process finished with exit code 0
```

Result

Thus, the **Travelling Salesman Problem** is successfully solved by finding the route that visits all cities exactly once and returns to the starting city with minimum total travel distance or cost.